

A Programmable, Multimode Operational 3U-VPX Based Digital Transceiver & Processing Module For CIT-MKXIIA IFF

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Abstract: This paper relates to the realization of 3U-VPX based programmable Digital Transceiver to support different modes (1, 2, 3, A, B, C, D, S and Mode5) and configurations of Identification of Friend and Foe (IFF) systems. L- Band signals are directly digitized and generated using high speed serializer (JESD204B) based RF-DAC and RF-ADC. Minimal and generic hardware with programmability of critical functions results in realizing a compact transceiver module which can be used in variants of IFF i.e. Interrogator, Transponder and CIT (Combined Interrogator and Transponder). This paper also discusses about the timing synchronization of ADC, DAC and FPGA with inbuilt processor using digitally programmable delays among clock sources. Module performs the role of signal processing and data processing as well for different modes of operation.

Keywords—Identification of friend and foe (IFF), Interrogator and transponder (CIT), Digital Transceiver, RF ADC and DAC, Mode S, Mode 5 and SoC, Minimum shift keying (MSK).

I. INTRODUCTION

Identification Friend-or-Foe (IFF) is a secure and encrypted aircraft identification system that tells user the difference between a friendly and hostile aircrafts. IFF systems use radio frequency (RF) devices and electromagnetic waves to communicate with the aircrafts for identifying friendly or foe in the airspace. This equipment sends the interrogation pulse to the aircraft and checking the information received are current password. As the advancement of the technology same system can be used for all the modes of communication to identify the foe aircraft.

An IFF system is a dual-channel transceiver system, using the 1030 MHz band for interrogating signals and the 1090 MHz band for response signals from the base station called as interrogator and revers as transponder as 1090MHz for interrogation and 1030 MHz for response. Operation can be divided into four modes, two for military use and two as common between military and civilian aircrafts. Mode 1 has 64 reply codes, and is used in military Air Traffic Control (ATC) to determine type and mission of aircraft. Mode 2 is reserved for military use. It has 4096 possible reply codes that is used to identify 'Tail Number' of an aircraft. Mode C is Automatic Altitude Reporting Mode. It returns the current altitude of the aircraft.

II. SYSTEM OVERVIEW

Antennae, diplexer, transceiver, digitization & waveform generation, Arinc & Milbus are the critical sub-modules of

IFF system [1]. Algorithms include phase matching among channels after digitizing [2], pulse-re-construction [3], aircraft identification, trackers and plots generation. Signal processing and data processing (SP&DP) module is heart of any Radar. Signal processing module includes FPGA based design to perform real time algorithms. Processor-based module suits in development of data processing with multi-core. The partial analyzed data from signal processing module interfaces with data processing module over high speed serial communication protocol. Standards of serial communication protocols are decided based on the processor and FPGA speed grade as well as system parameters.

Proposed dual channel programmable digital transceiver block diagram is shown in Fig 1.

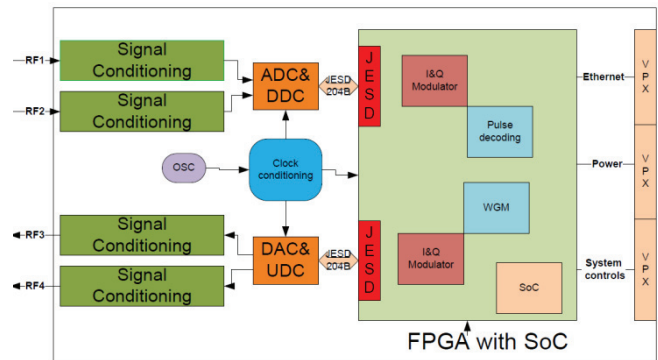


Fig. 1. Block diagram of 3U-VPX based Digital Transceiver & Signal Processing Module

Signals directly received from the antenna i.e. RF1 and RF2 are given to the module through diplexer. Signal conditioning block contains the amplification and filtering section to band limit and boost the signal to the ADC functioning level as shown in the Fig 1. Data converters i.e. RF ADC (Analog to Digital Converter) used in receive chain and RF DAC (Digital to Analog converter) used in transmit chain interfaces with SERDES protocol (JESD204B). JESD204B is a high-speed serial protocol which can transmit data upto 15GSPS drastically reduces the PCB design complexity and interfaces which helps in realizing compact system.

In the existing Radar system designs, architecture of soft processor within the FPGA has been used. This architecture adds bottleneck on speed of data processing. Signification development in VLSI technology has resulted in processor chips with inbuilt FPGA popularly known as HPS (Hard

Processor System). Architecture discussed in this paper uses single programmable device (HPS) for signal processing, data processing and interface of IFF Radar. Enhancing the purity of digitized data i.e. band limiting filters, frame analysis and generation of different modulation is performed within FPGA.

III. IFF MODES AND WAVEFORM GENERATION

Proposed digital receiver is dual channel transmitter and receiver module. Dual channel RF DAC is used to generate directly L- band signal which interfaces with FPGA using JESD204B high speed serial protocol as shown in Fig 2. Base band or IF (Intermediate Frequency) signal is provided to DAC for RF generation. Based on module application in interrogator or transponder configuration, user can select the frequency word to generate the 1090MHz signal or 1030MHz signal.

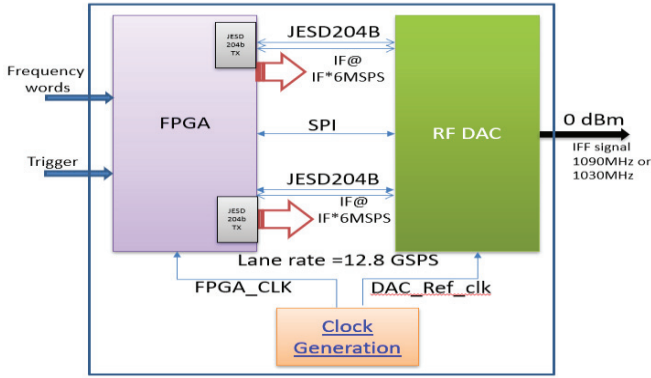


Fig. 2. Direct L-Band signal generation using RF DAC

A. Interrogation signals

The RF signal transmitted by the ground or base station is conventionally called an interrogation. Two pulses P1 and P3 are transmitted via the interrogate beam of the antenna and the spacing of these two pulses will determine the data content of the transponder reply. Fig 3 shows the characteristic of the interrogation signal.

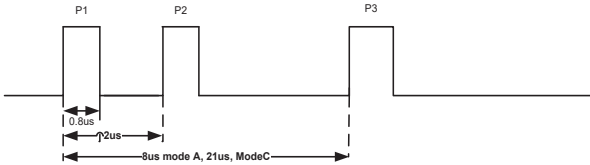


Fig. 3. Interrogation-signal format [4]

TABLE I. SPACING OF INTERROGATOR MODE PULSES [4]

Mode	P1-P3 pulse spacing(us)	User	Use
1	3	Military	Identification
2	5	Military	Identification
3/A	8	Civil/Military	Identity
C	21	Civil	Height

B. Transponder signals

According to the accepted standards of transponder of IFF received the signal on 1030MHz and reply with 1090MHz signal for the interrogation from the ground or base station. A Mode-S format frame (also called as squitter) consists of 112 or 56 data bits with a preceding preamble are shown in Fig 4.

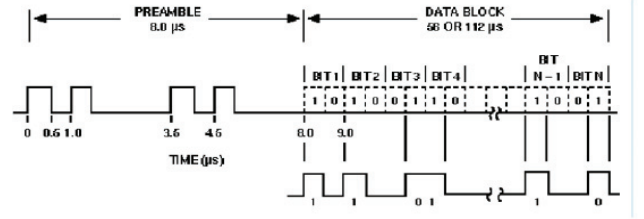


Fig. 4. Frame format of MODE-S mode [4]

Pulse modulated transmits data for the IFF Transponder is shown in the figure 5. For the generation of the below signal, a very high accurate and precise signal generator is required to meet the rise time and fall time of the transmitted signals.

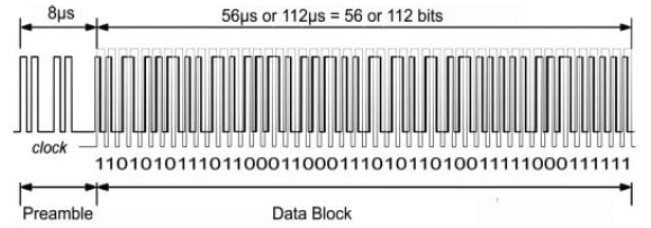


Fig. 5. Ideal transponder signal generation in base band [4]

RF chain in the signal conditioning block is design in such a way to meet the signal rise/fall time [8].

$$\text{Rise time } t_r = 0.35 / BW \quad (1)$$

Where BW is the bandwidth of the signal conditioning block filter bandwidth which directly affect the transmitted signal rise/fall time. As per standard rise time should be in the range of $\geq 0.05\mu s$ or $\leq 0.1\mu s$ [4].

Convention RF transceiver is having the pulse stretching issue so difficult to achieve very narrow pulse in order of 100ns. IFF system uses a very high-power amplifiers and with high power amplifiers, there is spectral regrowth [9] which can further increase the pulse stretching in transmitted signal. Due to the advancement in VLSI technology, digital transceivers are able to generate very narrow pulses with very good rise and fall time. Using proposed design, we have achieved the $\sim 35\text{ns}$ rise time and $\sim 80\text{ns}$ fall time as discussed in Results section.

IV. DIRECT SAMPLING RECEIVER AND PULSE DECODING

Pulse stretching is one of the major draws back for the conventional mixer-based RF receiver. Due to very narrow pulses used in the IFF system, it's very difficult for the pulse decoding. Using direct sampling digital receiver, above stated problems has been minimized. Received transponder signal is show in Fig 6. Due to minimal pulse stretching, pulse is recovered currently after pulse decoding algorithm.

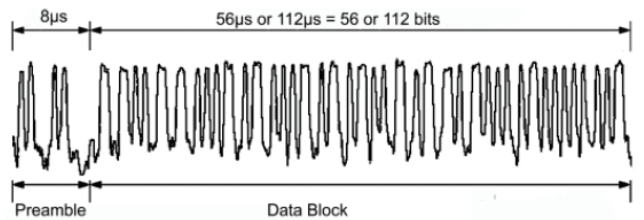


Fig. 6. Received transponder signal [4]

Function block diagram of the digital receiver is shown in Fig 7. For better sensitivity of digital receiver, signal has been amplified using LNA (Low noise amplifier) and given to the RF ADC. RF ADC is having inbuilt numerical control oscillator (NCO) which is used for IQ demodulation inside the ADC and directly baseband signal is transfer FPGA using JESD204B.

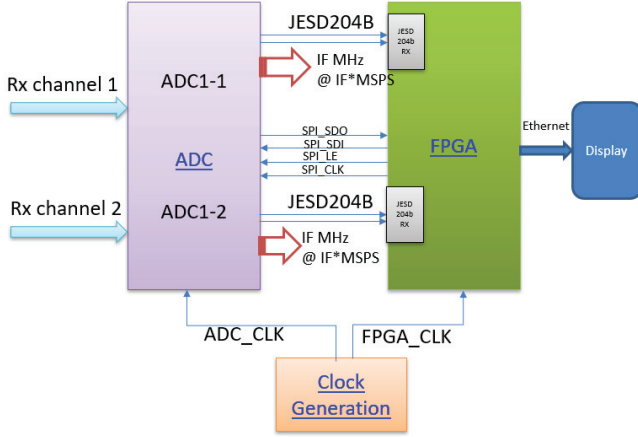


Fig. 7. Direct sampling digital receiver using RF ADC

V. TIMING SYNCHRONIZATION OF ADC,DAC AND FPGA BASED SoC

One of the major challenges of transceiver used in any RADAR system is timing synchronization across the multiple software as well as hardware blocks. Some of the systems required the synchronization in the order of 10ps between the channels. To achieve synchronization, in general is to use physically length matched cables. For any designer, it becomes a challenging task to match the path length for the clock between the different devices.

To overcome from the above mention problem, a programmable delay between multi-channel, phase locked loop (PLL) is used [5]. The proposed scheme is shown in Fig 8.

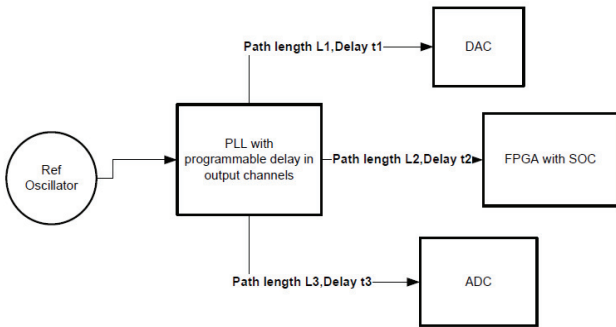


Fig. 8. Timing and synchronization with PLL programmable delay

As show in figure 8, due to the PCB length matching constrained is difficult to achieve the same path length. To match the path delay between the channel, programable delay in the outputs of PLL is used. Calibration has been done and finally 20ps synchronization between the DAC, ADC and FPGA has been achieved.

VI. IMPEMETATION

Minimum and generic hardware with programmability of critical functions results in realizing a compact and

multimode operational support for combined interrogator and transponder (CIT) system of IFF. Proposed hardware can be used for the both interrogator and transponder with change in firmware inside the FPGA. For the operation of interrogator, both the channel of the transmitter channels is generating 1030 MHz signal and for transponder 1090MHz with user selectable modes. Different modes are having different bandwidth of the signal. Designed digital receiver support maximum of 24 MHz signal bandwidth with Minimum shift keying (MSK) modulation in Mode 5.

Frequency switching and phase noise is one of the critical parameters for transmission. In legacy architectures [6], PLL and DDS combination is used. In the proposed architecture, these functionalities are performed by DAC only. Developed 3U-VPX based digital transceiver is shown in the Fig 9.

For operating as Interrogator, both the channels are receiving 1090MHz signals and for Transponder 1030MHz. Transmission is with reversed frequencies mentioned with user selectable modes and signal bandwidth. VPX connector is used to minimize the discrete connectors and it support the data transmission up to 15GSPS data rate. For the programmability of the module ethernet are provided using the same VPX connector.

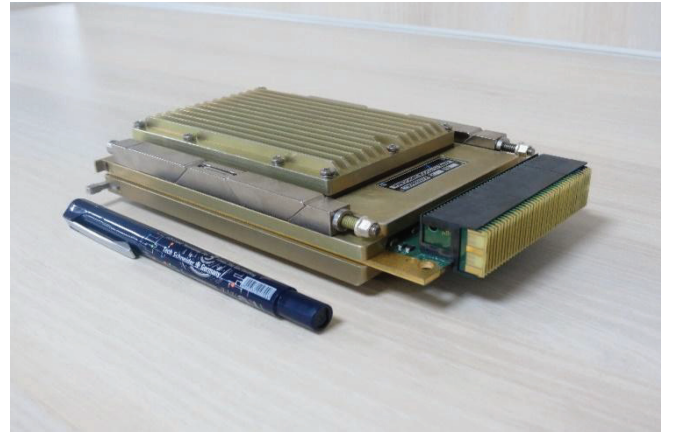


Fig. 9. Developed 3U-VPX based digital transceiver

VII. RESULTS

Modules are realized in 3U-VPX form-factors. Critical parameters are discussed in this section. Generation of L-Band signals using RF-DAC is characterized in spectrum analyzer as shown in Fig 10 and Fig 11. Received digitized

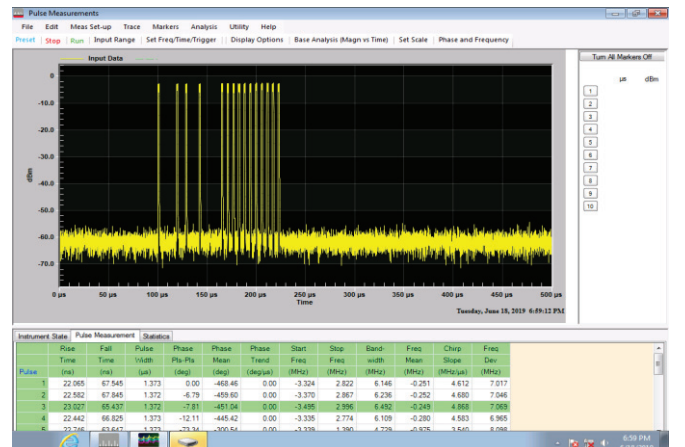


Fig. 10. Generation of Transmission Signals in L-Band



Fig. 11. Measurement of rise and fall time of transmitted signal

data at different power level is analyzed using HDC Pro (From Texas Instruments) as shown in Fig 12.

Achieved rise time is ~35ns and fall time is ~80 ns.

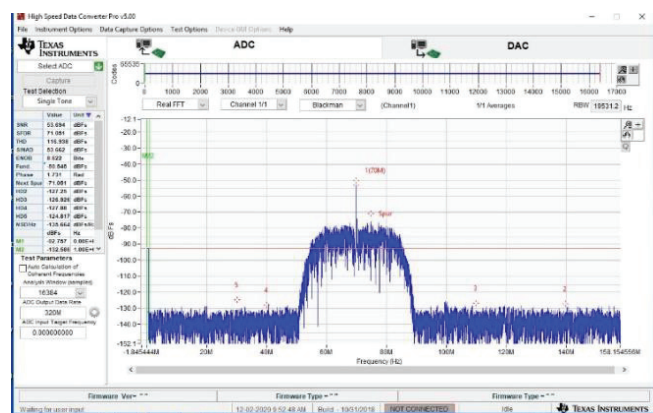


Fig. 12. Digitization received Signals in L-Band for -77dBm input [7]

VIII. ADVANTAGE AND FUTURE SCOPE OF WORK

Using the proposed programable digital transceiver, a very narrow pulse with programmable bandwidth can be generated up to L-band of the signal. Received path has been designed to receive the bandwidth up to 25 MHz which support all the modes of IFF. Due to the digital receiver, all the RF components are design in digital domain makes it very high-speed switching and minimum pulse starching receiver. Using the firmware changes, same hardware can to use in the different mode of IFF. Present work will be extended to make a generic hardware for a wide-band, high bandwidth and ultra-low phase noise programable digital transceiver which can work up to C-band of the signal and provide all the signal processing platform within the same transceiver.

IX. CONCLUSION

This paper covers the development of digital transceivers in industry standard 3U-VPX. The paper also covers the advantages by adapting the concept of Direct Digitalization and Direct Generation in design. The developed hardware can be used in combination for configurable IFF Radar platform with very less rise and fall time. Due to development of critical model in digital domain makes it a SWaP (size, weight and power) system.

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